

4.13.16

EE25BTECH11011 - Navya

Question:

The lines $p(p^2 + 1)x - y + q = 0$ and $(p^2 + 1)^2x + (p^2 + 1)y + 2q = 0$ are perpendicular to a common line for

- 1) exactly one value of p
- 2) exactly two values of p
- 3) more than two values of p
- 4) no value of p

Solution:

Given lines are,

$$p(p^2 + 1)x - y + q = 0 \quad (1)$$

$$(p^2 + 1)^2x + (p^2 + 1)y + 2q = 0 \quad (2)$$

The normal vectors of the lines are

$$\mathbf{n}_1 = \begin{pmatrix} p(p^2 + 1) \\ -1 \end{pmatrix}, \quad \mathbf{n}_2 = \begin{pmatrix} (p^2 + 1)^2 \\ p^2 + 1 \end{pmatrix} \quad (3)$$

If both lines are perpendicular to a common line with normal vector $\mathbf{m}$, then

$$\mathbf{n}_1^T \mathbf{m} = 0, \quad (4)$$

$$\mathbf{n}_2^T \mathbf{m} = 0. \quad (5)$$

This implies $\mathbf{m}$ lies in the null space of the matrix

$$N = \begin{pmatrix} \mathbf{n}_1^T \\ \mathbf{n}_2^T \end{pmatrix} = \begin{pmatrix} p(p^2 + 1) & -1 \\ (p^2 + 1)^2 & p^2 + 1 \end{pmatrix}. \quad (6)$$

For a non-trivial solution $\mathbf{m} \neq 0$ to exist,

$$\det(N) = 0. \quad (7)$$

Now,

$$\det = p(p^2 + 1)(p^2 + 1) - (-1)((p^2 + 1)^2) \quad (8)$$

$$= p(p^2 + 1)^2 + (p^2 + 1)^2 \quad (9)$$

$$= (p + 1)(p^2 + 1)^2 \quad (10)$$

Thus, the condition becomes

$$(p + 1)(p^2 + 1)^2 = 0 \implies p = -1 \quad \text{or} \quad p^2 + 1 = 0 \quad (11)$$

Since $p^2 + 1 = 0$ gives nonreal p , the only real solution is

$$\boxed{p = -1} \quad (12)$$

Hence there is exactly one real value of p .

